alligatorfood-code.r

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```r
setwd("~/Dropbox/Classes/536/CaseStudies/AligatorFood")

data <- read.table("alligatorfood-data.txt",header=TRUE)
attach(data)

#we need this library for the 'multinom' function
library(nnet)

y = cbind(Fish,Invertebrate,Reptile,Bird,Other)


(M1 = multinom(y ~ 1))
```

```
## # weights:  10 (4 variable)
## initial  value 352.466903
## final  value 302.181462
## converged
```

```
## Call:
## multinom(formula = y ~ 1)
##
## Coefficients:
##              (Intercept)
## Invertebrate  -0.4324211
## Reptile       -1.5988558
## Bird          -1.9783458
## Other         -1.0775589
##
## Residual Deviance: 604.3629
## AIC: 612.3629
```

```r
summary(M1)
```

```
## Call:
## multinom(formula = y ~ 1)
##
## Coefficients:
##              (Intercept)
## Invertebrate  -0.4324211
## Reptile       -1.5988558
## Bird          -1.9783458
## Other         -1.0775589
##
## Std. Errors:
##              (Intercept)
## Invertebrate   0.1644133
## Reptile        0.2515350
## Bird           0.2959078
```

```
## Other          0.2046663
##
## Residual Deviance: 604.3629
## AIC: 612.3629
```

```r
BIC(M1)
```

```
## [1] 625.9192
```

```r
(M2 = multinom(y ~ Lake))
```

```
## # weights:  25 (16 variable)
## initial  value 352.466903
## iter  10 value 281.030560
## iter  20 value 280.583926
## final  value 280.583844
## converged
```

```
## Call:
## multinom(formula = y ~ Lake)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford
## Invertebrate  -0.5008393  -1.5137909   0.55488981    0.8263598
## Reptile       -3.4962205   1.1937161   2.55175319    3.0107928
## Bird          -2.3982809   0.6065505  -0.49188808    1.2197770
## Other         -1.7048477   0.8686390  -0.08689071    1.4425750
##
## Residual Deviance: 561.1677
## AIC: 593.1677
```

```r
summary(M2)
```

```
## Call:
## multinom(formula = y ~ Lake)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford
## Invertebrate  -0.5008393  -1.5137909   0.55488981    0.8263598
## Reptile       -3.4962205   1.1937161   2.55175319    3.0107928
## Bird          -2.3982809   0.6065505  -0.49188808    1.2197770
## Other         -1.7048477   0.8686390  -0.08689071    1.4425750
##
## Std. Errors:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford
## Invertebrate   0.2833774   0.6029744    0.4341550    0.4612870
## Reptile        1.0148749   1.1817886    1.1083250    1.1099095
## Bird           0.6031161   0.7727128    1.1912598    0.8310587
## Other          0.4438217   0.5542879    0.7654142    0.6114775
##
## Residual Deviance: 561.1677
## AIC: 593.1677
```

```r
BIC(M2)
```

```
## [1] 647.3928
```

```r
(M3 = multinom(y ~ Gender))
```

```
## # weights:  15 (8 variable)
## initial  value 352.466903
## iter  10 value 301.192714
## final  value 301.129428
## converged

## Call:
## multinom(formula = y ~ Gender)
##
## Coefficients:
##              (Intercept) GenderMale
## Invertebrate  -0.2231478 -0.3578812
## Reptile       -1.7635569  0.2509570
## Bird          -1.7635851 -0.3680370
## Other         -0.9162928 -0.2708745
##
## Residual Deviance: 602.2589
## AIC: 618.2589
```

```r
summary(M3)
```

```
## Call:
## multinom(formula = y ~ Gender)
##
## Coefficients:
##              (Intercept) GenderMale
## Invertebrate  -0.2231478 -0.3578812
## Reptile       -1.7635569  0.2509570
## Bird          -1.7635851 -0.3680370
## Other         -0.9162928 -0.2708745
##
## Std. Errors:
##              (Intercept) GenderMale
## Invertebrate   0.2535467  0.3339731
## Reptile        0.4418517  0.5376858
## Bird           0.4418570  0.5958549
## Other          0.3162281  0.4153372
##
## Residual Deviance: 602.2589
## AIC: 618.2589
```

```r
BIC(M3)
```

```
## [1] 645.3714
```

```r
(M4 = multinom(y ~ Size))
```

```
## # weights:  15 (8 variable)
## initial  value 352.466903
## iter  10 value 294.689203
## final  value 294.606678
## converged

## Call:
## multinom(formula = y ~ Size)
```

```
##
## Coefficients:
##              (Intercept)       Size
## Invertebrate -0.08515712 -0.9489002
## Reptile      -2.10000556  0.8582954
## Bird         -2.28237308  0.5551345
## Other        -0.94734781 -0.2943686
##
## Residual Deviance: 589.2134
## AIC: 605.2134
```

```r
summary(M4)
```

```
## Call:
## multinom(formula = y ~ Size)
##
## Coefficients:
##              (Intercept)       Size
## Invertebrate -0.08515712 -0.9489002
## Reptile      -2.10000556  0.8582954
## Bird         -2.28237308  0.5551345
## Other        -0.94734781 -0.2943686
##
## Std. Errors:
##              (Intercept)      Size
## Invertebrate   0.2064721 0.3568642
## Reptile        0.4325127 0.5349882
## Bird           0.4694765 0.6063273
## Other          0.2702565 0.4149509
##
## Residual Deviance: 589.2134
## AIC: 605.2134
```

```r
BIC(M4)
```

```
## [1] 632.3259
```

```r
(M5 = multinom(y ~ Lake + Gender))
```

```
## # weights:  30 (20 variable)
## initial  value 352.466903
## iter  10 value 279.421076
## iter  20 value 277.733691
## final  value 277.732884
## converged
```

```
## Call:
## multinom(formula = y ~ Lake + Gender)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford GenderMale
## Invertebrate  0.01048504  -1.7605713   0.59314476    0.9609816 -0.8577619
## Reptile      -3.37256725   1.1400408   2.55949903    3.0361432 -0.1834060
## Bird         -2.12320607   0.4818741  -0.47393004    1.2811363 -0.4245404
## Other        -1.51431307   0.7842964  -0.07500933    1.4829188 -0.2871253
##
## Residual Deviance: 555.4658
```

```
## AIC: 595.4658
```

```r
summary(M5)
```

```
## Call:
## multinom(formula = y ~ Lake + Gender)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford GenderMale
## Invertebrate  0.01048504  -1.7605713   0.59314476    0.9609816 -0.8577619
## Reptile      -3.37256725   1.1400408   2.55949903    3.0361432 -0.1834060
## Bird         -2.12320607   0.4818741  -0.47393004    1.2811363 -0.4245404
## Other        -1.51431307   0.7842964  -0.07500933    1.4829188 -0.2871253
##
## Std. Errors:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford GenderMale
## Invertebrate   0.3584792   0.6187324    0.4425069    0.4722548  0.3694164
## Reptile        1.0863463   1.1945082    1.1088625    1.1134113  0.5873043
## Bird           0.7212115   0.7960695    1.1922056    0.8381776  0.6476502
## Other          0.5318941   0.5701319    0.7661014    0.6158038  0.4542938
##
## Residual Deviance: 555.4658
## AIC: 595.4658
```

```r
BIC(M5)
```

```
## [1] 663.2472
```

```r
(M6 = multinom(y ~ Lake + Size))
```

```
## # weights:  30 (20 variable)
## initial  value 352.466903
## iter  10 value 270.844027
## iter  20 value 270.041364
## final  value 270.040140
## converged
```

```
## Call:
## multinom(formula = y ~ Lake + Size)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford       Size
## Invertebrate -0.09083394  -1.6583241  0.937163858     1.121984 -1.4581521
## Reptile      -3.66588368   1.2431622  2.458817408     2.935296  0.3513836
## Bird         -2.72380722   0.6952142 -0.653348590     1.087850  0.6307298
## Other        -1.57283851   0.8262891  0.005522604     1.516419 -0.3313740
##
## Residual Deviance: 540.0803
## AIC: 580.0803
```

```r
summary(M6)
```

```
## Call:
## multinom(formula = y ~ Lake + Size)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford       Size
```

```
## Invertebrate -0.09083394  -1.6583241  0.937163858     1.121984 -1.4581521
## Reptile      -3.66588368   1.2431622  2.458817408     2.935296  0.3513836
## Bird         -2.72380722   0.6952142 -0.653348590     1.087850  0.6307298
## Other        -1.57283851   0.8262891  0.005522604     1.516419 -0.3313740
##
## Std. Errors:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford      Size
## Invertebrate   0.3080369   0.6128764    0.4718994    0.4905118 0.3959420
## Reptile        1.0589837   1.1853979    1.1181345    1.1164171 0.5800193
## Bird           0.7104046   0.7812686    1.2021440    0.8416698 0.6424808
## Other          0.4748331   0.5575480    0.7765972    0.6214385 0.4482525
##
## Residual Deviance: 540.0803
## AIC: 580.0803
```

```r
BIC(M6)
```

```
## [1] 647.8617
```

```r
(M7 = multinom(y ~ Gender + Size))
```

```
## # weights:  20 (12 variable)
## initial  value 352.466903
## iter  10 value 294.854746
## final  value 294.091727
## converged
```

```
## Call:
## multinom(formula = y ~ Gender + Size)
##
## Coefficients:
##              (Intercept)  GenderMale       Size
## Invertebrate -0.04280353 -0.09004186 -0.9211347
## Reptile      -2.08500394 -0.03135665  0.8677287
## Bird         -2.03235739 -0.61461112  0.7532196
## Other        -0.85769580 -0.19616474 -0.2331107
##
## Residual Deviance: 588.1835
## AIC: 612.1835
```

```r
summary(M7)
```

```
## Call:
## multinom(formula = y ~ Gender + Size)
##
## Coefficients:
##              (Intercept)  GenderMale       Size
## Invertebrate -0.04280353 -0.09004186 -0.9211347
## Reptile      -2.08500394 -0.03135665  0.8677287
## Bird         -2.03235739 -0.61461112  0.7532196
## Other        -0.85769580 -0.19616474 -0.2331107
##
## Std. Errors:
##              (Intercept) GenderMale      Size
## Invertebrate   0.2650229  0.3538189 0.3730316
## Reptile        0.5100510  0.5694016 0.5629408
## Bird           0.5204014  0.6337808 0.6439212
```

```
## Other          0.3336578  0.4378947 0.4373976
##
## Residual Deviance: 588.1835
## AIC: 612.1835
```

```r
BIC(M7)
```

```
## [1] 652.8523
```

```r
(M8 = multinom(y ~ Lake+Gender+Size))
```

```
## # weights:  35 (24 variable)
## initial  value 352.466903
## iter  10 value 271.274792
## iter  20 value 268.935538
## final  value 268.932740
## converged

## Call:
## multinom(formula = y ~ Lake + Gender + Size)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford GenderMale
## Invertebrate   0.1690702  -1.7805555   0.91304120     1.155722 -0.4629388
## Reptile       -3.4161432   1.1296426   2.53024945     3.061087 -0.6276217
## Bird          -2.4321397   0.5754699  -0.55020075     1.237216 -0.6064035
## Other         -1.4309095   0.7667093   0.02603021     1.557820 -0.2524299
##                    Size
## Invertebrate -1.3361658
## Reptile       0.5571846
## Bird          0.7300740
## Other        -0.2905697
##
## Residual Deviance: 537.8655
## AIC: 585.8655
```

```r
summary(M8)
```

```
## Call:
## multinom(formula = y ~ Lake + Gender + Size)
##
## Coefficients:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford GenderMale
## Invertebrate   0.1690702  -1.7805555   0.91304120     1.155722 -0.4629388
## Reptile       -3.4161432   1.1296426   2.53024945     3.061087 -0.6276217
## Bird          -2.4321397   0.5754699  -0.55020075     1.237216 -0.6064035
## Other         -1.4309095   0.7667093   0.02603021     1.557820 -0.2524299
##                    Size
## Invertebrate -1.3361658
## Reptile       0.5571846
## Bird          0.7300740
## Other        -0.2905697
##
## Std. Errors:
##              (Intercept) LakeHancock LakeOklawaha LakeTrafford GenderMale
## Invertebrate   0.3787475   0.6232075    0.4761068    0.4927795  0.3955162
## Reptile        1.0851582   1.1928075    1.1221413    1.1297557  0.6852750
```

```
## Bird           0.7706720   0.7952303    1.2098680    0.8661052  0.6888385
## Other          0.5381162   0.5685673    0.7777958    0.6256868  0.4663546
##                     Size
## Invertebrate 0.4111827
## Reptile      0.6466092
## Bird         0.6522657
## Other        0.4599317
##
## Residual Deviance: 537.8655
## AIC: 585.8655
```

```r
BIC(M8)
```

```
## [1] 667.2032
```

```r
#LRT
1-pchisq(M7$deviance-M8$deviance,length(coef(M8))-length(coef(M7)))
```

```
## [1] 1.228388e-06
```